

Geometry

Unit 3

Lesson 1 Practice

Name **Answer Key** _____

Date _____

Period _____

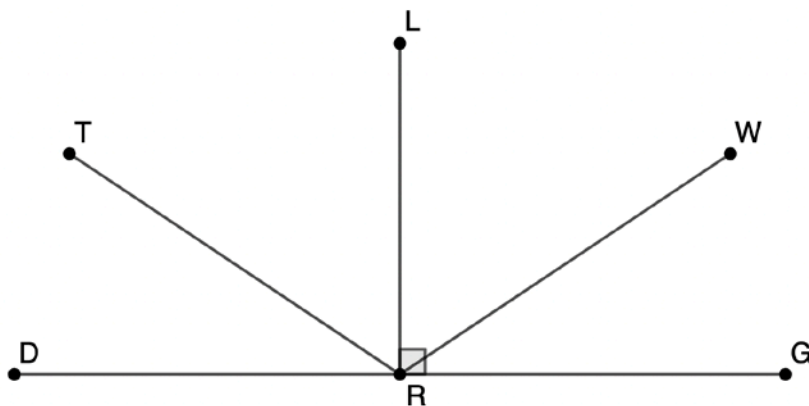
Use the phrases in the word bank to complete the proof. Some phrases may be used more than once.

Word Bank

Definition of congruence	Linear Pair Postulate
Supplements of congruent angles are congruent.	Complements of congruent angles are congruent.
Angle addition postulate	Substitution property of equality
Definition of supplementary angles	Definition of complementary angles

Given: $\angle DRT \cong \angle GRW$, $m\angle DRL = 90^\circ$, $m\angle GRL = 90^\circ$

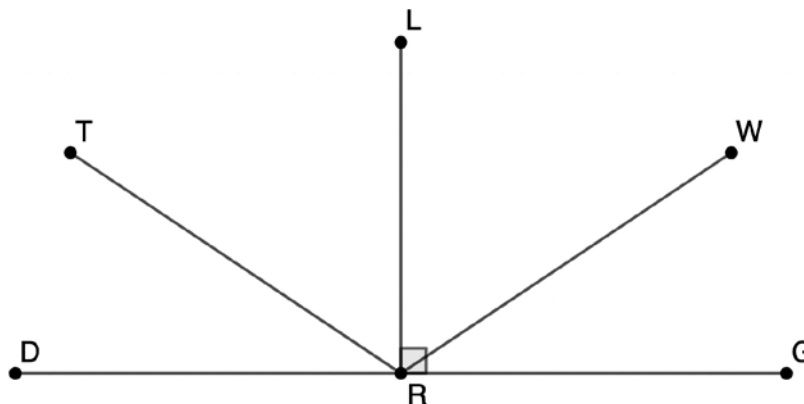
Prove: $\angle LRT \cong \angle LRW$



Complete the table using the word bank to prove $\angle LRT \cong \angle LRW$.

Statement	Reason
$m\angle DRL = 90^\circ$	Given
$m\angle LRG = 90^\circ$	Given
$m\angle DRT + m\angle LRT = m\angle DRL$	Angle addition postulate
$m\angle DRT + m\angle LRT = 90^\circ$	1. Substitution property of equality
$\angle DRT$ and $\angle LRT$ are complementary angles	2. Definition of complementary angles
$m\angle LRW + m\angle GRW = m\angle LRG$	Angle addition postulate
$m\angle LRW + m\angle GRW = 90^\circ$	3. Substitution property of equality
$\angle LRW$ and $\angle GRW$ are complementary angles	4. Definition of complementary angles
$\angle DRT \cong \angle GRW$	Given
$\angle LRT \cong \angle LRW$	5. Complements of congruent angles are congruent

Lines $\overline{DG}$, $\overline{TR}$, $\overline{LR}$, and $\overline{RW}$ are shown where $\angle DRT \cong \angle GRW$, $m\angle DRL = 90^\circ$, $m\angle GRL = 90^\circ$ and $m\angle DRT = \left(\frac{5}{2}w - 8\right)^\circ$ and $m\angle GRW = (2w - 1)^\circ$.



6. Find the value of w .

$$w = 14$$

7. What is $m\angle DRT$?

$$m\angle DRT = 27^\circ$$

8. What is $m\angle DRW$?

$$m\angle DRW = 153^\circ$$

9. What is $m\angle LRW$?

$$m\angle LRW = 63^\circ$$

10. What is $m\angle TRG$?

$$m\angle TRG = 153^\circ$$